Deep Learning for Natural Language Processing

Today's goal • 3/3 items

Module 1
Feedforward Neural Nets and Recurrent Neural Networks

Introduction to the Course
- Course Updates and Accessibility Support — Reading • 1 min
- Earn Academic Credit for Your Work! — Reading • 10 min
- Course Support — Reading • 10 min
- Assessment Expectations — Reading • 5 min
- AI Citation and Acknowledgement — Reading • 10 min

Feedforward Networks
- The Perceptron — Video • 6 min
- The XOR Problem — Video • 4 min
- Feedforward Networks — Video • 7 min
- Feedforward Networks in NLP — Video • 8 min
- Mathematics of Feedforward Networks — Reading • 25 min
- Sentence Embeddings with Feedforward Networks — Video • 5 min
- Feedforward Networks — Practice Assignment • 15 min

Recurrent Neural Networks
- Sentence Embeddings with Recurrent Neural Networks — Video • 4 min
- Sequence Labelling with Recurrent Neural Networks — Video • 4 min
- Mathematics of Recurrent Neural Networks — Reading • 25 min
- LSTMs and GRUs — Video • 6 min
- Bidirectional Recurrent Neural Networks — Video • 3 min
- Hierarchical Recurrent Neural Networks — Video • 5 min
- Recurrent Neural Networks in Python — Video • 8 min
- Recurrent Neural Networks — Practice Assignment • 15 min

Implementing and Training Neural Networks
- Loss Functions — Video • 6 min
- The Intuition behind Gradient Descent — Video • 5 min
- Stochastic Gradient Descent — Video • 6 min
- Stochastic Gradient Descent — Reading • 25 min
- Stochastic Gradient Descent in Python — Video • 6 min
- Stochastic Gradient Descent — Lab • 30 min
- Training Neural Networks — Practice Assignment • 15 min

Review and Exercises
- AI Policy Quiz — Graded Assignment • 5 min
- Feedforward Networks, Recurrent Neural Networks, and How to Train Them — Graded Assignment • 15 min
- Recurrent Neural Networks for Sequence Labelling — Programming Assignment • 1h 30m

Module 2
Sequence to Sequence Models, Attention, Transformers

Module 3
Transfer Learning

Module 4
Large Language Models

Assessment Expectations

0:16 / 5:38 2x

Auto-graded Assessments

As you complete any auto-graded assessment, please consider the following:

- Some assessments may include a set of instructions with a list of resources you are and aren't to use. While these assessments may not be proctored, please consider adhering to this list of resources as the assessment was designed with the list in mind.
- Most auto-graded assessments will allow for multiple attempts. Between attempts, it is recommend you review feedback provided for correct and incorrect responses and relevant course materials before starting your next attempt.
- Some programming assignments will contain hidden test cases that are graded. If you do not receive full credit for your submission, please review your assignment summary for the passed and failed tests so you can further optimize your solutions.

Artificial Intelligence (AI) Usage Guidance

Artificial intelligence tools—software that reproduces text, images, computer code, audio, video, and other content—have become widely available. Well-known examples include ChatGPT for text and DALL•E for images. This statement governs all such tools, including those released during your enrolled session.

It is your responsibility—not the tool's—to assure the quality, integrity, and accuracy of work you submit in all CU Boulder courses. Keep in mind that AI tools are designed to create content that seems to have been produced by a human, not to produce accurate or reliable content; therefore, relying on an AI tool may result in inaccurate content. Be wary of unintentional plagiarism or data fabrication. Act with integrity, for the sake of both your personal character and your academic record.

Consider the impact of using AI tools on your learning goals: in most cases, using an AI-generated answer without critical review and significant input from your end is contrary to the overall goal of personal learning and development in your chosen area.

Respect copyright and privacy. You may only enter your own data into AI tools. Do not enter private, sensitive, or copyrighted data from others into AI tools without their consent. Avoid using AI for purposes that are private, sensitive, or high-risk.

Consider your environmental impact. Using AI models requires large amounts of electricity, water, and rare earth minerals. When using AI, keep in mind there are environmental concerns associated with its use.

Artificial Intelligence (AI) Usage Allowed by this Course

The following table indicates each assessment in this course and the AI usage allowed for that assessment.

Note: If you are taking this course for credit or transfer prior work into a for-credit version of this course, you will be held to this standard.

Assessment	AI Usage Policy
4 Auto-Graded Quizzes	Limited
4 Programming Assignments	Limited
Final Exam	Limited

For assessments that allow Limited AI Usage

You may use AI for planning, idea development and research. Your final submission must be generated by you, showing how you have developed and refined these ideas. You must critically evaluate and modify any AI generated content you use. If you use AI tools on assignments in this class, document your usage according to the AI Usage Documentation Guidelines below.

You CAN use AI tools to...

- **Check your spelling and grammar**
- **Ask for help with programming syntax and general programming questions that are not directly related to the solution of the assignment in question.**
- **Generate project ideas, outlines, brainstorming.**

You CANNOT use AI tools to...

- **Submit exact or paraphrased assignment questions** or prompts into the AI tool
- **Generate answers** for graded assignments in place of your own work or thinking
- **Help with a graded assignment in any way without documenting usage** (see *AI Usage Documentation Guidelines* below)

AI Usage Documentation Best Practices

When using AI tools in your academic work, it is essential to accurately document how and why you used them. AI tools should be considered as aids rather than primary sources. Therefore, remember to:

- **Reference AI assistance.** In your references or bibliography section, mention the AI tool used and the nature of its contribution. For example: "Assistance provided by [AI Tool Name] for generating initial content ideas, response to [prompt], [month, day, year]".
- **Document your process.** Save screenshots, dialog, prompt, queries or version history to document your process if you used AI. Include documentation in the submission if the assignment format allows. Otherwise, save it in your system in case you are asked by a program or university staff.
- Check this course's reading on AI Citation and Acknowledgement for further guidance on expectations for this course.

Assessments and Academic Honesty

As you complete assessments, you are expected to perform to the highest standards of academic honesty. Please keep the following in mind:

- Copying the provided rubrics to use as your own work is considered plagiarism.
- Copying your peers work that you have reviewed in peer review assignments is considered plagiarism.
- Using generative AI services, such as ChatGPT, to write your peer review assignments or answer other graded assignment questions and submitting those as your own work is considered plagiarism.
- Using question-and-answer websites, such as Stack Overflow or Chegg, to post assessment questions and solicit solutions and then submitting responses as your own work is considered plagiarism.

Go to next item ✓ Completed

Like Dislike Report an issue

Go to next item →